

In Problems 1 to 12, differentiate from first principles.

1. $y = x$

2. $y = 7x$

3. $y = 4x^2$

4. $y = 5x^3$

5. $y = -2x^2 + 3x - 12$

6. $y = 23$

7. $f(x) = 9x$

8. $f(x) = \frac{2x}{3}$

9. $f(x) = 9x^2$

10. $f(x) = -7x^3$

11. $f(x) = x^2 + 15x - 4$

12. $f(x) = 4$

13. Determine $\frac{d}{dx}(4x^3 - 1)$ from first principles

14. Find $\frac{d}{dx}(3x^2 + 5)$ from first principles

1. (a) $5x^5$ (b) $2.4x^{3.5}$ (c) $\frac{1}{x}$
2. (a) $\frac{-4}{x^2}$ (b) 6 (c) $2x$
3. (a) $2\sqrt{x}$ (b) $3\sqrt[3]{x^5}$ (c) $\frac{4}{\sqrt{x}}$
4. (a) $\frac{-3}{\sqrt[3]{x}}$ (b) $(x-1)^2$ (c) $2\sin 3x$
5. (a) $-4\cos 2x$ (b) $2e^{6x}$ (c) $\frac{3}{e^{5x}}$
6. (a) $4\ln 9x$ (b) $\frac{e^x - e^{-x}}{2}$ (c) $\frac{1 - \sqrt{x}}{x}$
7. Find the gradient of the curve $y = 2t^4 + 3t^3 - t + 4$ at the points (0, 4) and (1, 8)
8. Find the co-ordinates of the point on the graph $y = 5x^2 - 3x + 1$ where the gradient is 2
9. (a) Differentiate $y = \frac{2}{\theta^2} + 2\ln 2\theta - 2(\cos 5\theta + 3\sin 2\theta) - \frac{2}{e^{3\theta}}$